

Tuong Luu

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EDUCATION

University of Waterloo

Bachelor of Mathematics, Honours Computational Mathematics

Waterloo, Ontario

Expected May 2028

EXPERIENCE

Software Engineer Intern

Carta

Kitchener, Ontario

May. 2026 – Aug. 2026

- Saved **\$50K annually** by replacing a third-party API documentation platform with a code-first Docusaurus site, including an automated content pipeline and API gateway routing
- Replaced manual customer provisioning with self-service Snowflake token authentication across Django, React, and Terraform, using feature flags to control rollout
- Diagnosed upstream data gaps and API gateway routing errors behind recurring on-call pages; reduced rows flagged by production tests from **255K+ to 2.5K**
- Fixed a framework-level matcher bug causing intermittent false failures in an internal LLM evaluation framework; added **30+ regression tests** for AI agent skills

Software Developer Intern

Lynkr Inc

Toronto, Ontario

Sep. 2025 – Dec. 2025

- Built tenant isolation and RBAC for a FastAPI/Next.js AI automation platform that connects business systems and orchestrates workflows across them, **serving 500+ beta users**.
- Reduced GKE deployment time by **40%** by moving database seeding to a Kubernetes job
- Implemented multi-factor authentication using TOTP and Twilio, plus JWT session controls to secure account access
- Added audit logging for user actions, administrative changes, and security events to support activity tracing

PROJECTS

AniDis – Anime Discovery Platform | TypeScript, Next.js, FastAPI, PostgreSQL

- Built a full-stack platform for anime discovery, watch tracking, and community discussions, with verified accounts, OAuth account linking, and API rate limits
- Designed a transactional-outbox pipeline to deliver in-app notifications and email through separate services, decoupling delivery from API requests
- Automated releases with versioned Docker images and preview environments with isolated databases; managed DNS and access policies as code with OpenTofu

Kyzen – 2D Game Engine | Java, LWJGL, OpenGL

- Designed a batch renderer for scenes with **1,000+ objects per frame**; integrated texture atlases to reduce draw calls by **over 70%**
- Implemented an entity-component system to compose game objects from reusable components

TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, Bash

Frameworks: Django, FastAPI, React, Next.js

Infrastructure & Tools: Docker, Kubernetes, Terraform/OpenTofu, Git, Linux

Databases: PostgreSQL, MongoDB, Snowflake